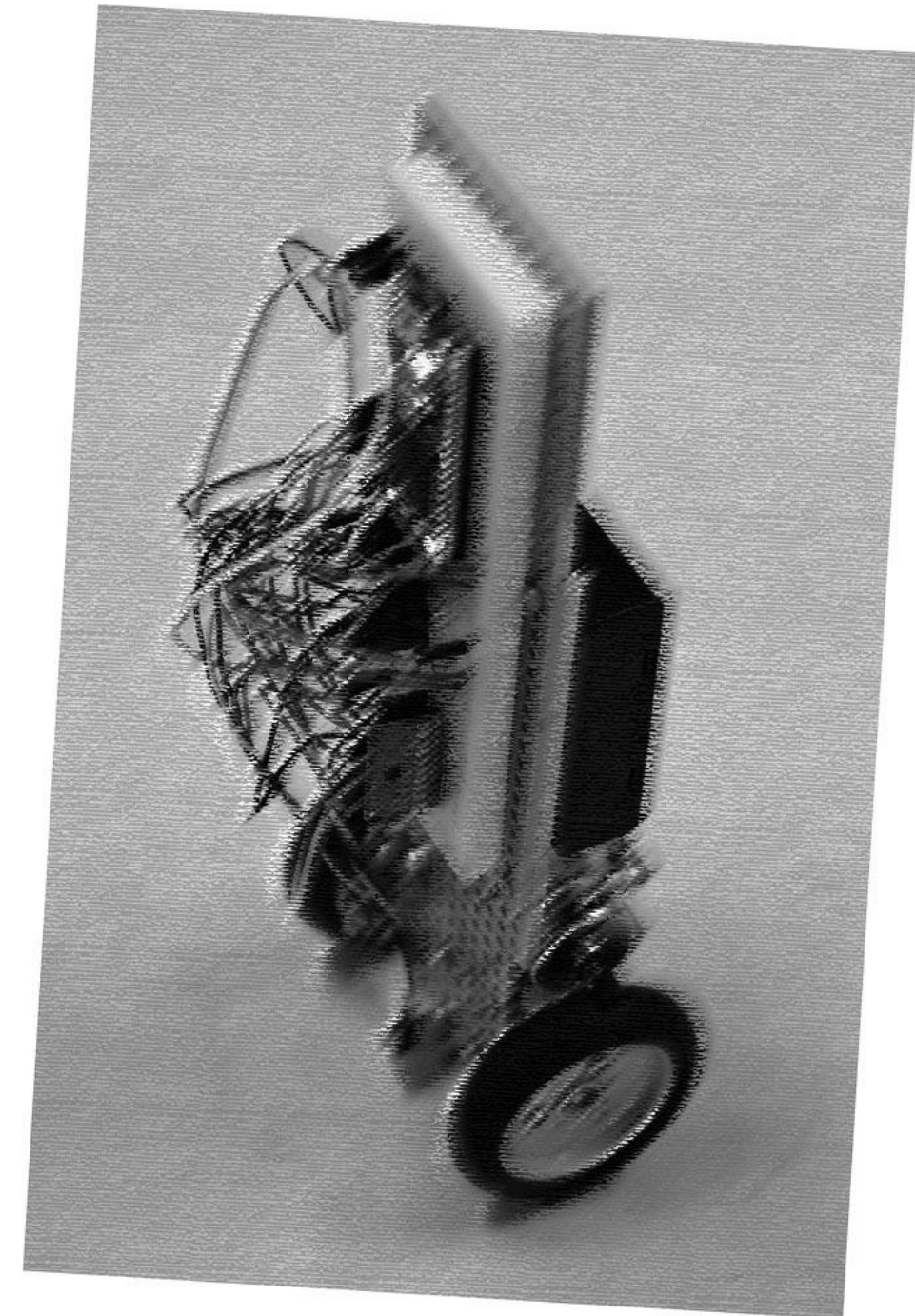


PRACTICAL EXERCISE · WEEK 1

# Construction of a Wheeled Inverted Pendulum

Shaoqi (シャオ チー)

**COURSE** Department of Precision Engineering · 2026 S1S2  
**SESSIONS** Mondays, 13:00–18:00 · Room 326, Engineering Bldg. 14  
**TODAY** Course Introduction — 10 minutes



# Week 0 - You Should Have Done These.

✓ Register a Keil Studio Cloud Account

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✓ Install Processing 4

---

✓ Install Fritzing

---

✓ Download the Course Material

No worries if you don't get it done.

THE FOUR-WEEK ARC

# Timeline

1

Week 1

Standing

Assembly & bring-up  
Angle-only **standing** demo

2

Week 2

Position control

Build a rotary encoder  
**Position control**  
Simulation in Simulink

3

Week 3

EXTENSION

Load & disturbance tests  
Advanced control (LQR/MPC)  
.....

4

Week 4

presentation

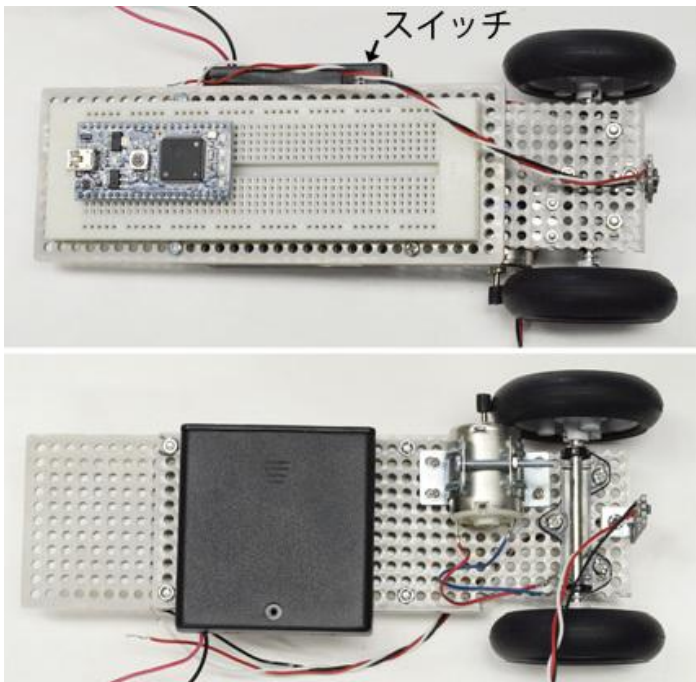
10-minute presentation  
Live demo & Q&A  
Final grade · 40%

# Step 0 - Work in parallel.

Assign the tasks among yourselves. If there are four people in the group, two people can work on one task together. Alternatively, the fourth member can begin circuit assembly.

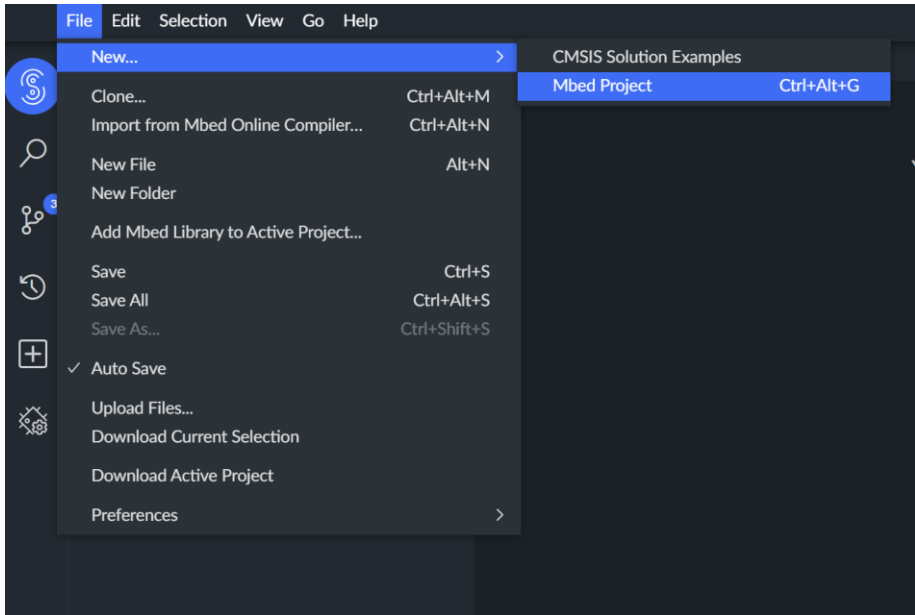
**A TASK A**  
STUDENT A · MECHANICAL

**Mechanical assembly**



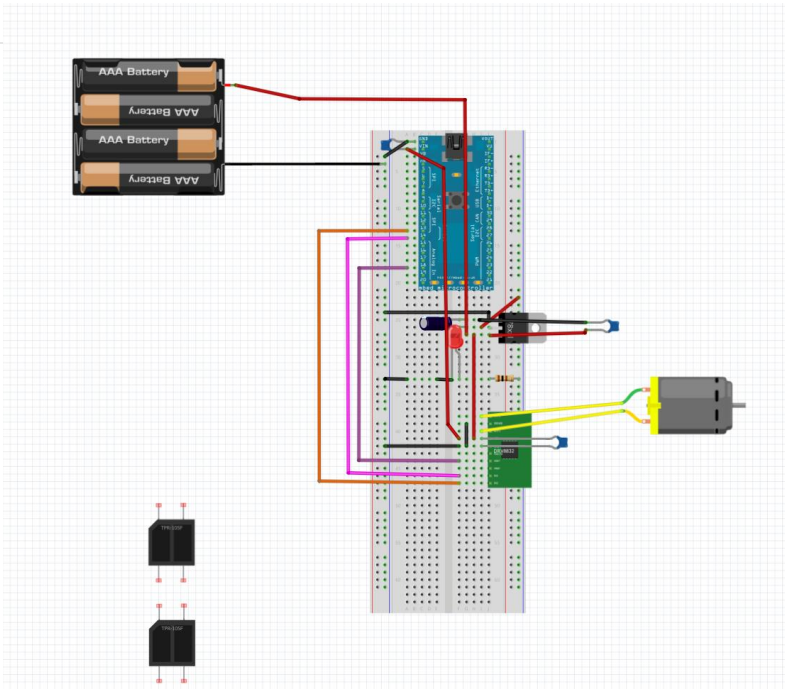
**B TASK B**  
STUDENT B · FIRMWARE

**Microcontroller**



**C TASK C**  
STUDENT C · WIRING DESIGN

**Fritzing breadboard layout**

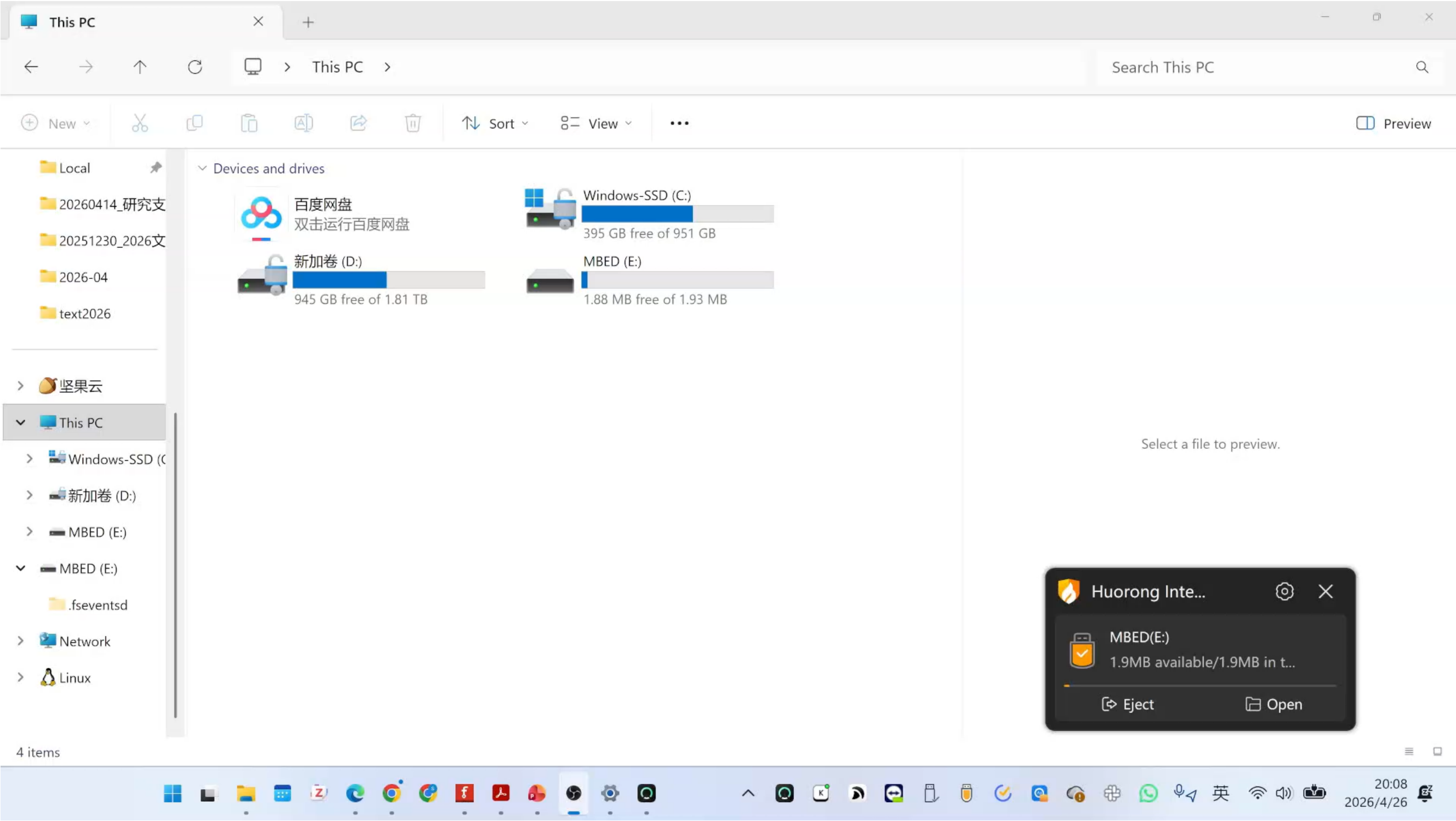


# Microcontroller

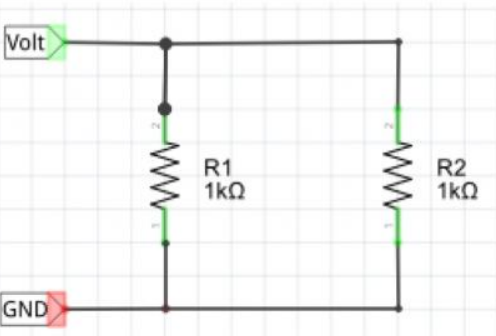
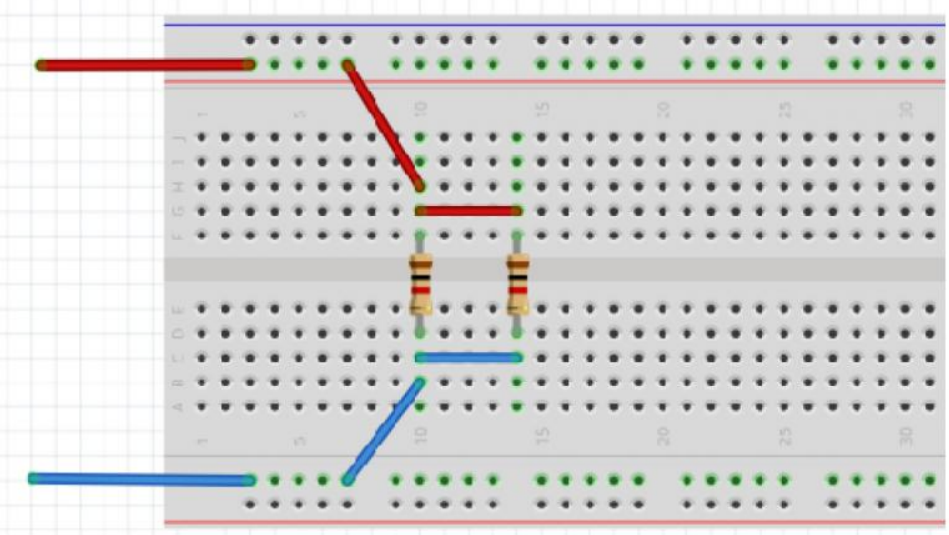
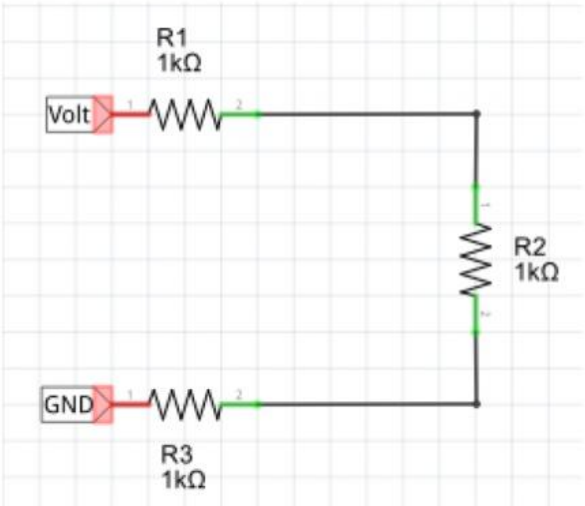
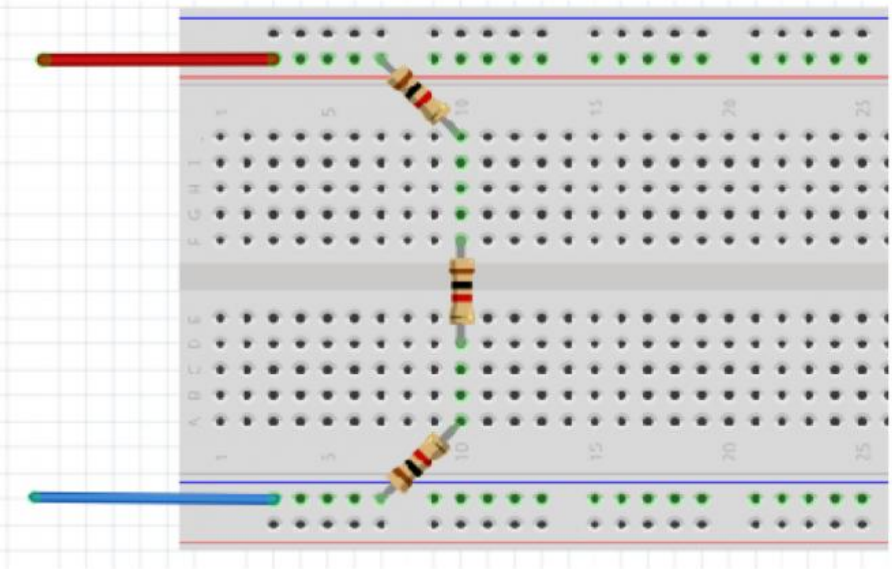
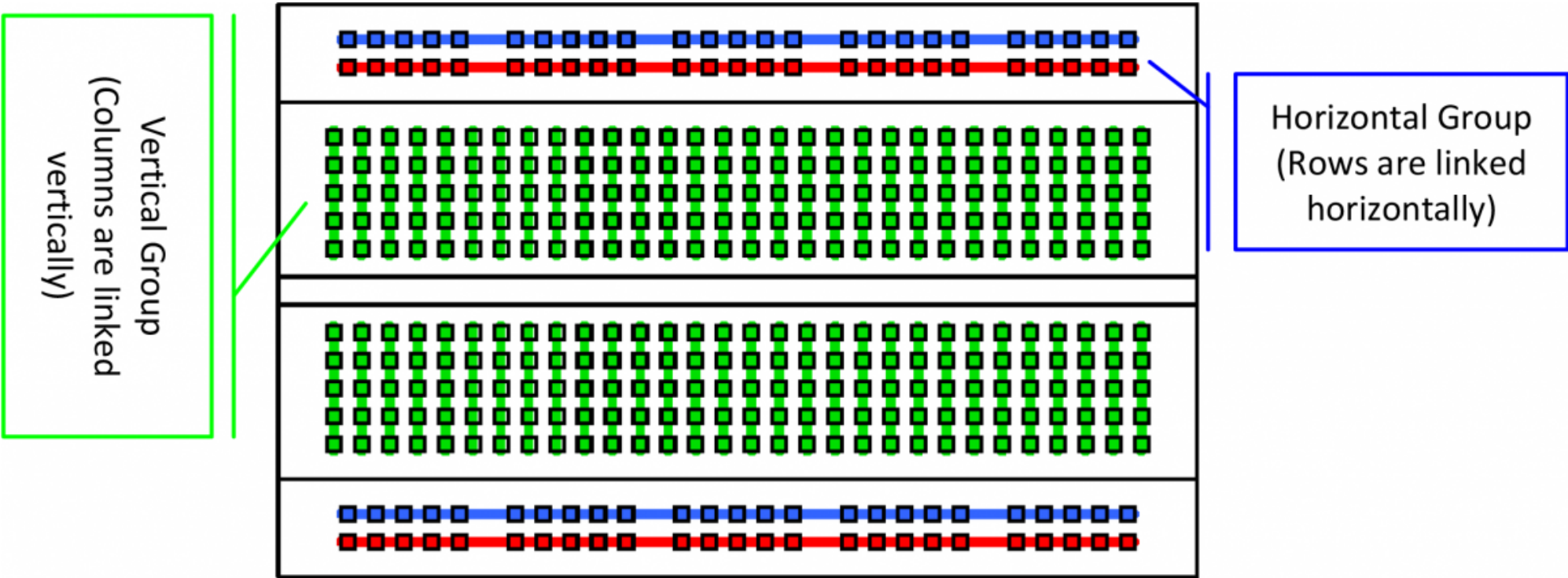
course\_materials\debug\_tools

-  **D1\_motor\_controller.bin** **compiled binary**  
generated by Keil Cloud → Copy to Microcontroller
-  **D1\_motor\_controller.cpp** **source code**  
Open with Notepad → Copy to Keil Cloud
-  **D1\_motor\_controller.pde** **debugging tool**

# Microcontroller

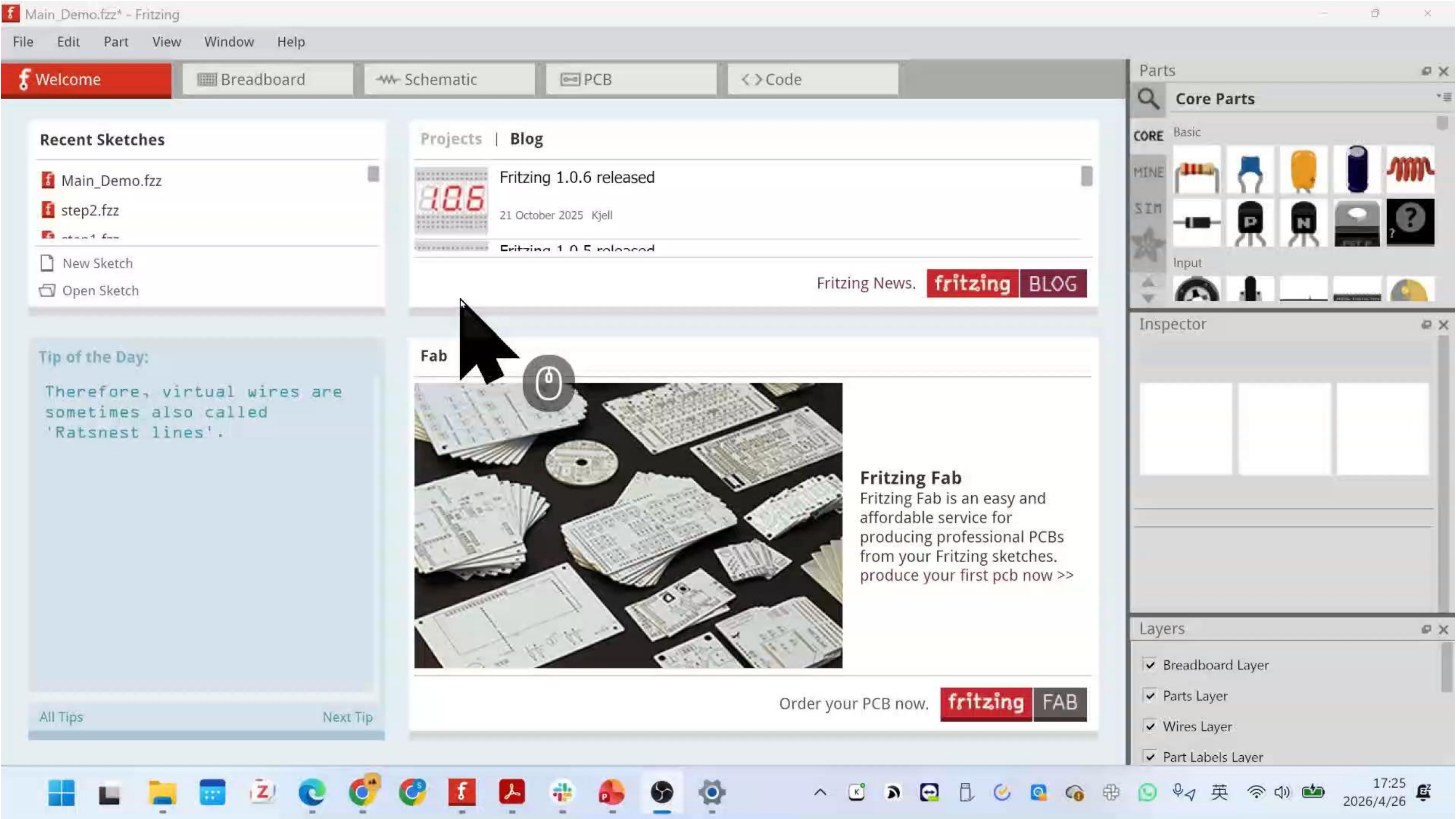


# Breadboard



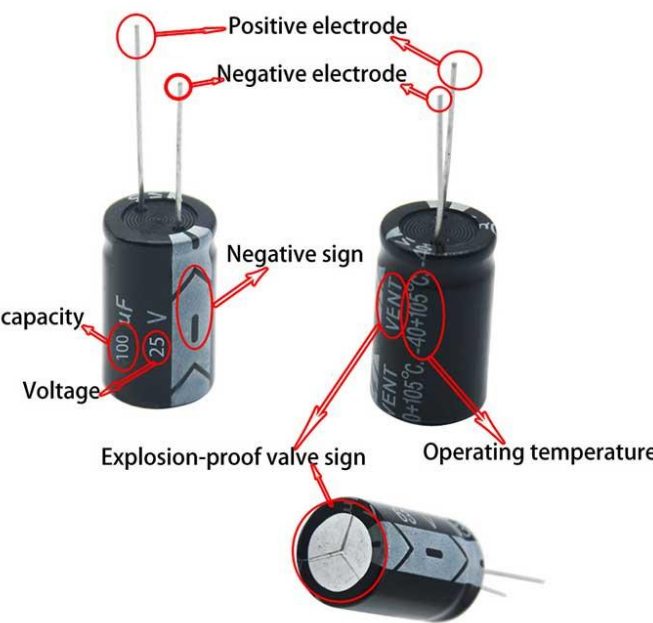
It's quite abstract.

# Fritzing-assisted breadboard layout (optional)

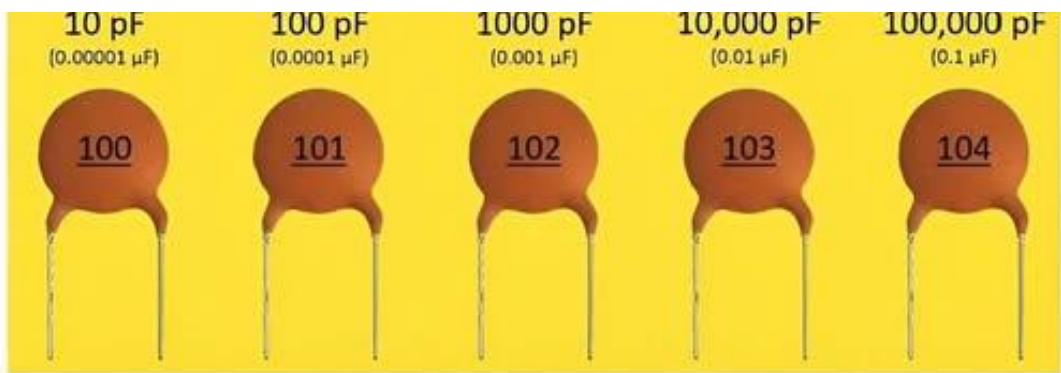


# Component values and polarity

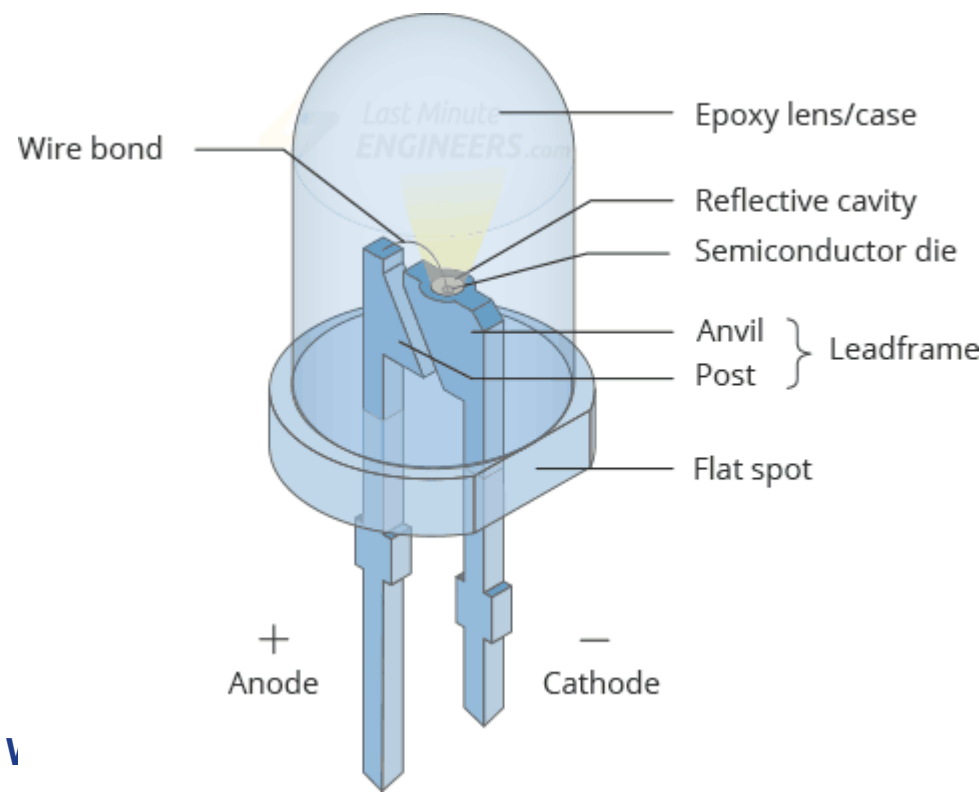
NONPOLARIZED      POLARIZED



$104 = 10 \times 10^4 = 100,000 \text{ pF} = 0.1 \mu\text{F}$



|        |     |       |           |
|--------|-----|-------|-----------|
| 0.1µF  | 104 | 100nF | 100,000pF |
| 0.12µF | 124 | 120nF | 120,000pF |
| 0.15µF | 154 | 150nF | 150,000pF |
| 0.22µF | 224 | 220nF | 220,000pF |

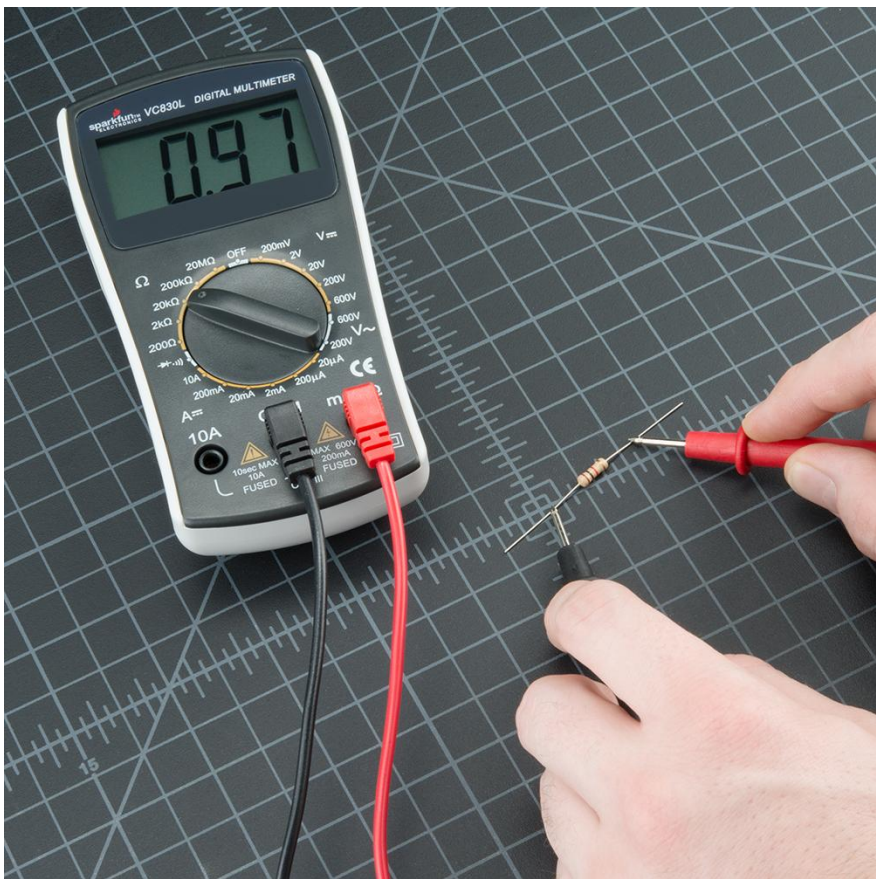


Example resistor: 1 6 0 x1k = 160kΩ

| Color  | 1st | 2nd | x <sup>th</sup> | Multiplier | Tolerance |
|--------|-----|-----|-----------------|------------|-----------|
| Black  | 0   | 0   | 0               | x1         |           |
| Brown  | 1   | 1   | 1               | x10        | ±1%       |
| Red    | 2   | 2   | 2               | x100       | ±2%       |
| Orange | 3   | 3   | 3               | x1k        | ±3%       |
| Yellow | 4   | 4   | 4               | x10k       | ±4%       |
| Green  | 5   | 5   | 5               | x100k      | ±0.5%     |
| Blue   | 6   | 6   | 6               | x1M        | ±0.25%    |
| Purple | 7   | 7   | 7               | x10M       | ±0.1%     |
| Gray   | 8   | 8   | 8               | x100M      |           |
| White  | 9   | 9   | 9               | x1,000M    |           |
| Gold   |     |     |                 | x0.1       | ±5%       |
| Silver |     |     |                 | x0.01      | ±10%      |

Common Tolerances

Reading Axial Resistor Codes



# Team Collaboration (3 STEP)

TOOL

115200

motor\_controller

Motor Control (mbed: F <0..100>, B <0..100>, S)

Port: (no ports) (click to cycle)

Refresh

Connect

Forward (F)

Backward (B)

Stop (S)

Duty (%): 30

Drag sets duty. On mouse release, sends F/B <duty> if motor is running.

Expected IO State (UI-side)

DIR = STOP

p13(motor\_f) = 0

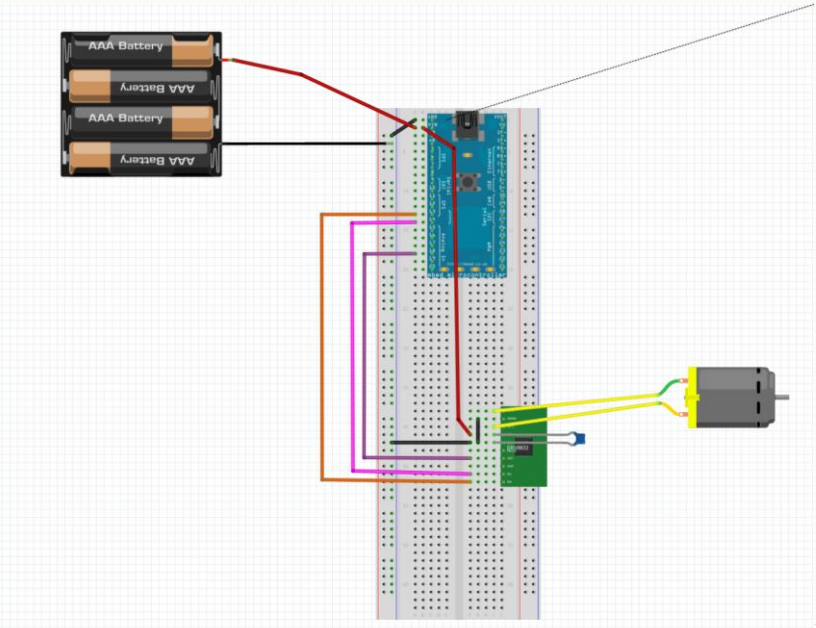
p14(motor\_r) = 0

p18(motor\_v duty) = 30%

Note: shown state is what UI sent; mbed does not send real IO feedback.

Log: Ready.

## STEP 1 Motor Controller



TOOL

115200

analog\_input\_visualizer

4ch Analog Debug (Frame: AA 55 CH1 CH2 CH3 CH4)

Port: COM3 (click to cycle)

Refresh

Disconnect

Decoded Channels (raw 0..255) [CH1=p16, CH2=p17, CH3=p19, CH4=p20]

|               |               |                |               |
|---------------|---------------|----------------|---------------|
| CH1 (p16): 79 | CH2 (p17): 62 | CH3 (p19): 151 | CH4 (p20): 10 |
| CH1: 0.310    | CH2: 0.243    | CH3: 0.592     | CH4: 0.039    |

Frames: 897 Bad bytes: 0 (auto re-sync by header)

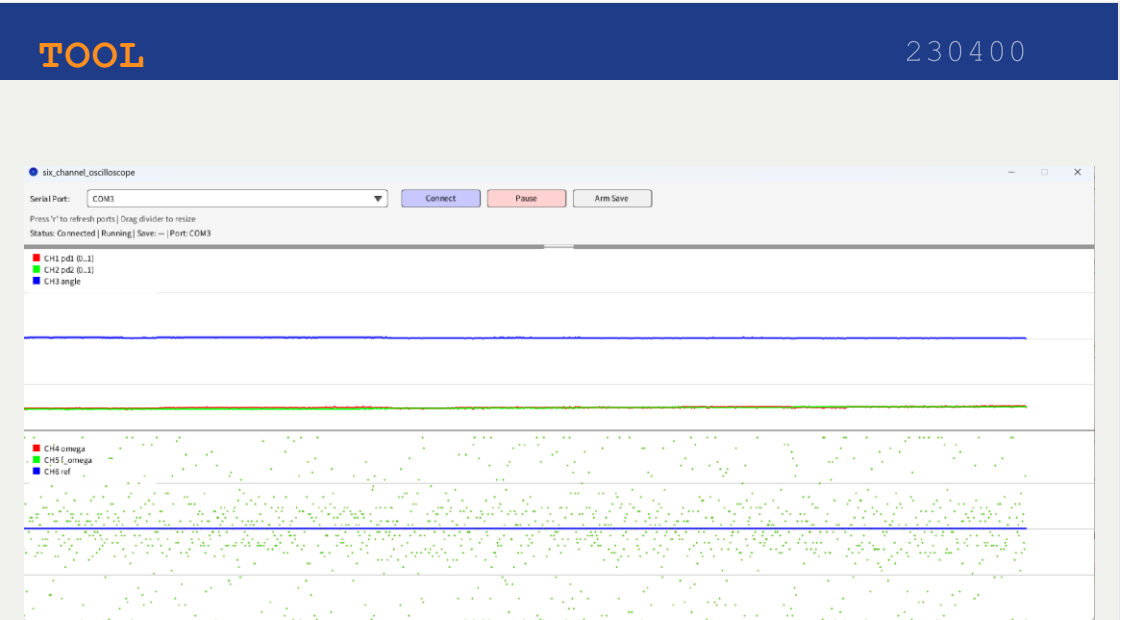
Simple Bar View (0..1)

|         |             |       |
|---------|-------------|-------|
| CH1 p16 | <div></div> | 0.310 |
| CH2 p17 | <div></div> | 0.243 |
| CH3 p19 | <div></div> | 0.592 |
| CH4 p20 | <div></div> | 0.039 |

Tip: if values look frozen -> check baud / wiring / device is sending frames.

Log: Connected: COM3 @ 115200

## STEP 2 Analog Input Visualizer



## STEP 3 · Standing Example

AI tools may be used for editing and refinement.

# Step 1 – Motor Controller

TOOL115200

motor\_controller

Motor Control (mbed: F<0..100>, B<0..100>, S)

Port: (no ports) (click to cycle)

RefreshConnect

Forward (F)Backward (B)Stop (S)

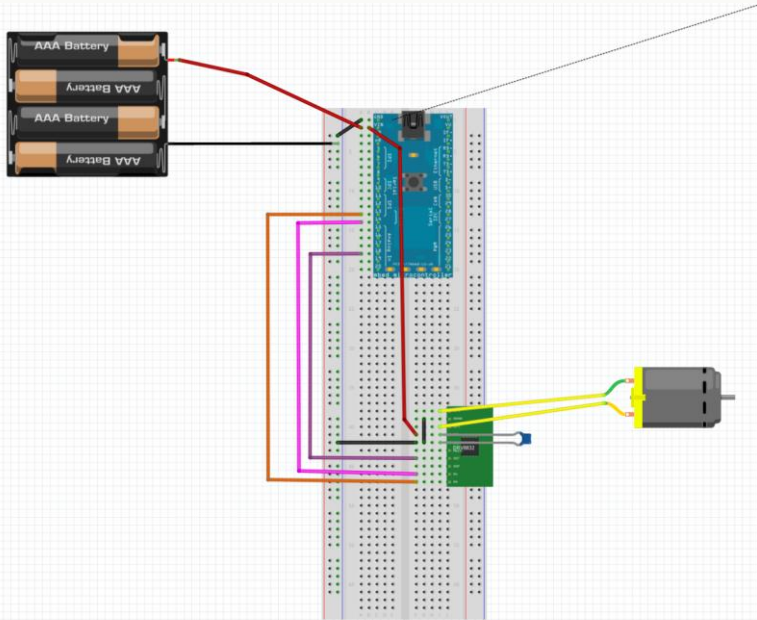
Duty (%): 30

Drag sets duty. On mouse release, sends F/B <duty> if motor is running.

Expected IO State (UI-side)  
DIR = STOP  
p13(motor\_f) = 0p14(motor\_r) = 0p18(motor\_v duty) = 30%  
Note: shown state is what UI sent; mbed does not send real IO feedback.

Log: Ready.

## STEP 1 Motor Controller



D1\_motor\_controller

text2026 > course\_materials > debug\_tools > D1\_motor\_controller

Search D1\_motor\_controller

New > > Sort > View > Preview

| Name                    | Date modified  | Type               | Size  |
|-------------------------|----------------|--------------------|-------|
| D1_motor_controller.bin | 2026/1/3 16:45 | BIN File           | 28 KB |
| D1_motor_controller.cpp | 2026/1/3 16:49 | CPP File           | 2 KB  |
| D1_motor_controller.pde | 2026/1/3 17:05 | PDE File           | 8 KB  |
| README.md               | 2026/2/4 13:05 | Markdown File      | 7 KB  |
| README.pdf              | 2026/2/4 13:05 | Adobe Acrobat 文... | 49 KB |

Documents

Pictures

Music

Videos

Local

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This PC

fritzing and 3 more tabs -...

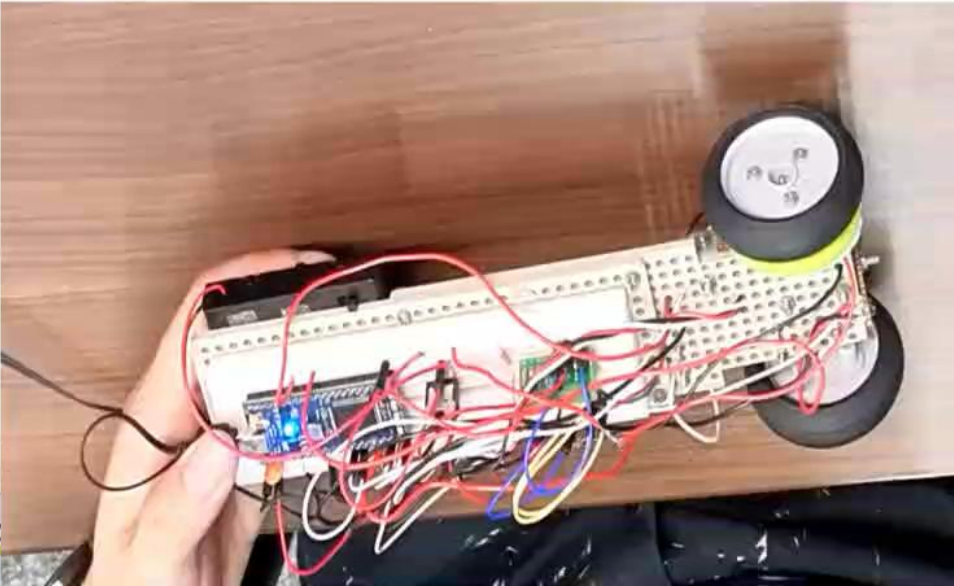
D1\_motor\_controller...

Downloads - File Explorer

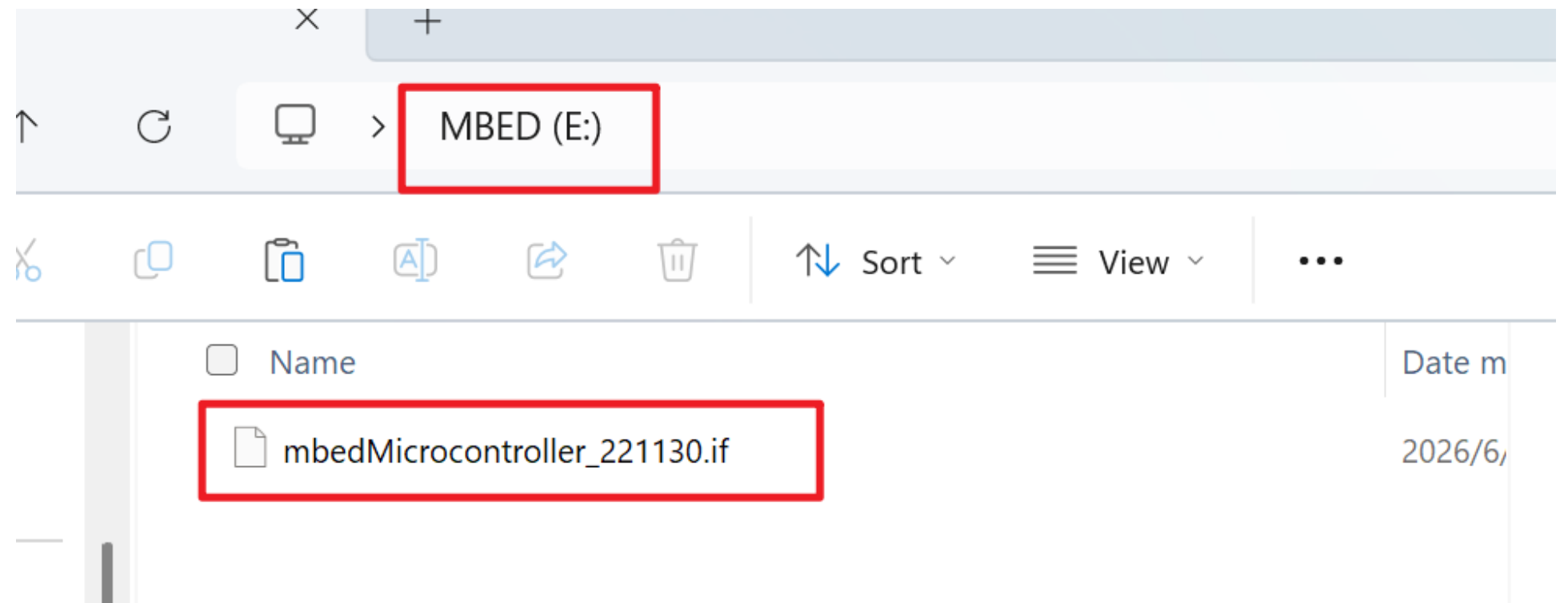
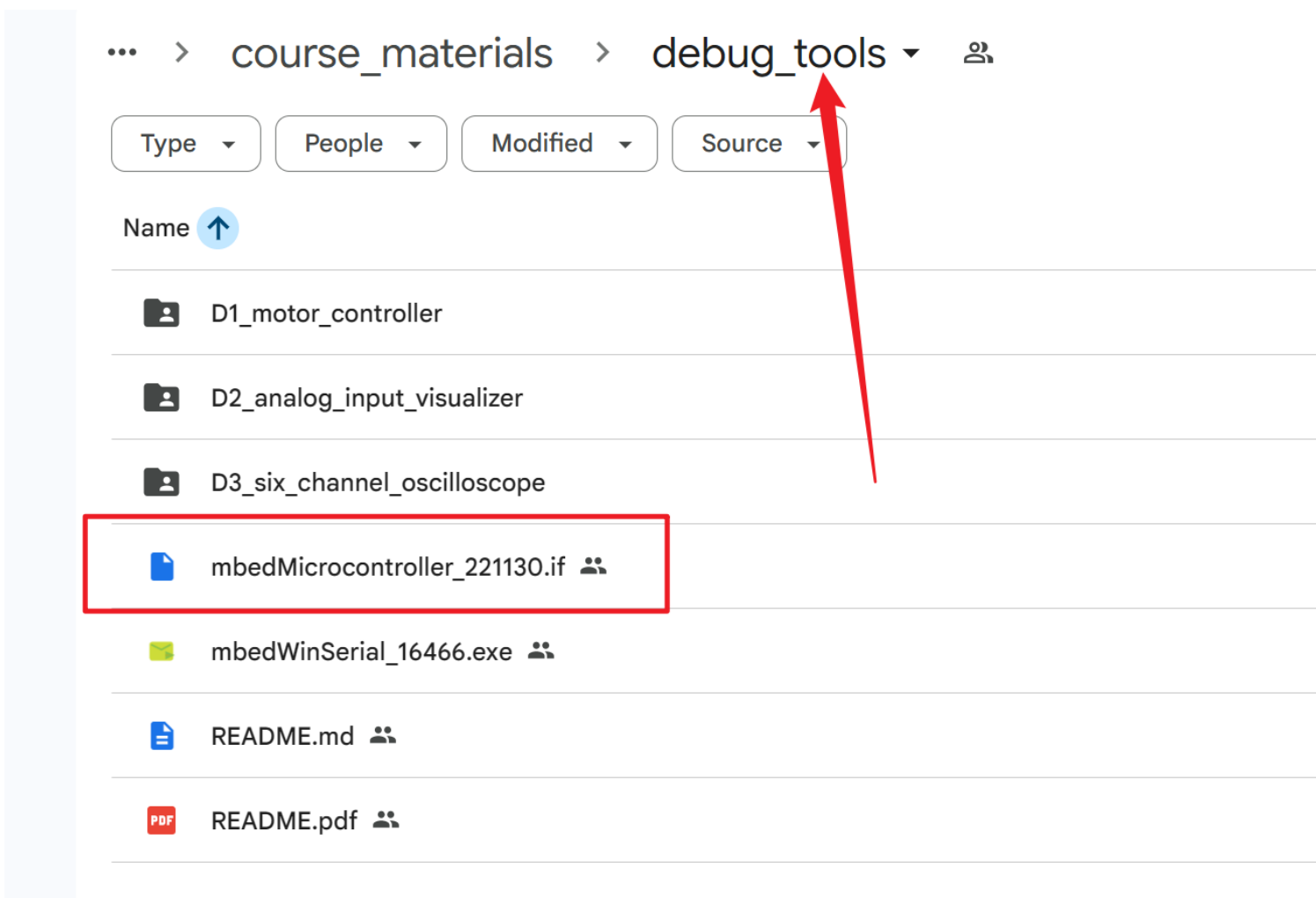
MBED (E:) - File

5 items 1 item selected 1.75 KB

The file you are attempting to preview could harm your computer. If you trust the file and the source you received it from, open it to view its contents.



**If you are unable to connect to the Windows serial port, you can try updating the mbed firmware.**



# Grading Criteria

Weekly submissions · 60%

Final Presentation · 40%

| Week | Deliverables                                                                                                                   | Score (%)                      | Notes                                                                                                                                                                 |
|------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| W1   | — One video: standing demo                                                                                                     | 60%<br>(Weeks 1–3 submissions) | Due by <b>9 pm</b> on the day of the session. Late until <b>9 pm the following day</b> ; the <b>maximum score is 50%</b> .<br>→ See Week 1 submission details (p. 24) |
| W2   | — Video 1: real hardware<br>— Video 2: simulation without disturbance<br>— Video 3: simulation with disturbance                |                                | Due by <b>9 pm</b> on the day of the session. Late until <b>9 pm the following day</b> ; the <b>maximum score is 50%</b> .<br>→ See Week 2 submission details (p. 43) |
| W3   | — Video 1: maximum disturbance/load on real hardware<br>— Video 2: maximum disturbance in simulation<br>— One-page PDF summary |                                | Due by <b>9 pm</b> on the day of the session. Late until <b>9 pm the following day</b> ; the <b>maximum score is 50%</b> .<br>→ See Week 3 submission details (p. 45) |
| W4   | Final presentation & Q&A                                                                                                       | 40%                            | Live 10-minute group presentation; no written submission required.<br>→ See Week 4 details (p. 47)                                                                    |

DEADLINE

21:00

on the day of the class

Late submissions accepted until **21:00 the following day** . Late score capped at 50%.

Presentation in English

# Week 1 Submission

**Video (max 200 MB):**  
**show the wheeled inverted pendulum**  
**balancing upright for at least 5 seconds.**

Format      MP4 · smartphone recording recommended

Submit to    forms.gle/5huiVLJAj5Bh16Dj7

Week1\_GroupX.zip

If the robot cannot yet stand, record its current state and clearly show which parts are already working.

**Please also attach a PDF explaining the current status and the planned solution.**

Each group only needs one submission.

## DEADLINE

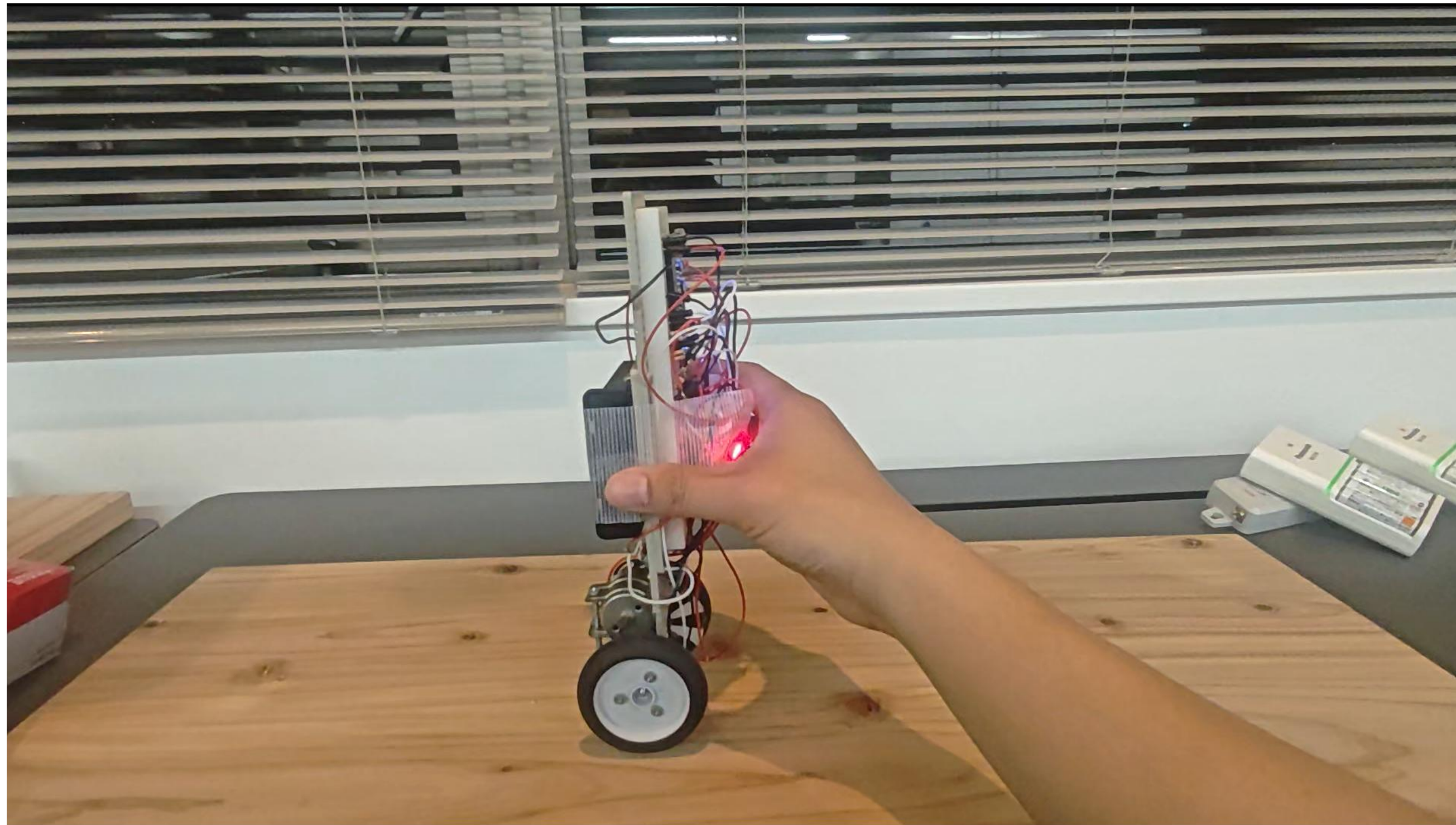
# 21:00

**on the day of the class**

Today — Monday, June 29

Late submissions accepted until **21:00 the following day** . Late score capped at 50%.

# Week 1 Submission



# Reminders

- If any parts are missing from the kit, please contact the TA.
- Some components in the kit may be damaged, so please test them by yourself.
- After flashing the .bin file, please restart the MCU.
- Remember to charge the battery.

UTokyo-Control-Practice-2026 / fritzing /

UTokyo2026 Add Fritzing demo files and guide videos (4-6); rem

| Name          |
|---------------|
| ..            |
| DRV8832.fzpz  |
| Main.fzz      |
| Main_Demo.fzz |
| Step1.fzz     |
| Step2.fzz     |
| TPR-105F.fzpz |
| mbed.fzpz     |

UTokyo-Control-Practice-2026 / video /

UTokyo2026 Add Fritzing demo files and guide videos (4-

| Name                      |
|---------------------------|
| ..                        |
| 1_mechanical_assembly.mp4 |
| 2_circuit_assembly.mp4    |
| 3_debugging_tuning.mp4    |
| 4_keil_cloud_guide.mp4    |
| 5_fritzing_guide.mp4      |
| 6_debug_tool_guide.mp4    |